INTERPRETING AND STEERING PROTEIN LANGUAGE MODELS THROUGH SPARSE AUTOENCODERS

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ABSTRACT

The rapid advancements in transformer-based language models have revolutionized natural language processing, yet understanding the internal mechanisms of these models remains a significant challenge. This paper explores the application of sparse autoencoders (SAE) to interpret the internal representations of protein language models, specifically focusing on the ESM-2 8M parameter model. By performing a statistical analysis on each latent component's relevance to distinct protein annotations, we identify potential interpretations linked to various protein characteristics, including transmembrane regions, binding sites, and specialized motifs. We then leverage these insights to guide sequence generation, shortlisting the relevant latent components that can steer the model towards desired targets such as zinc finger domains. This work contributes to the emerging field of mechanistic interpretability in biological sequence models, offering new perspectives on model steering for sequence design.

1 INTRODUCTION

Since the introduction of the transformer architecture (Vaswani, 2017), the capabilities of neural networks to model and generate natural language have increased dramatically. Yet, due to their black-box nature, we still lack a clear understanding of how these models achieve such capabilities (Rai et al., 2024). Recently, the mechanistic interpretability approach has been proposed, where researchers try to reverse engineer neural networks in a way similar to reverse engineering computer programs (Olah, 2022; Rai et al., 2024). This involves understanding which features the network is learning from our input data, and then how it performs operations with this set of features.

It has been observed that neural networks tend to encode high-level features as linear directions in their representation space—such as the gender direction in word embeddings (Park et al., 2023). Additionally, these models can store more facts and features than their parameter counts would suggest, a phenomenon known as superposition (Elhage et al., 2022). This phenomenon represents a core problem for interpretability: as a single neuron activation can be polysemantic and represent multiple features simultaneously.

Recently, sparse autoencoders (SAEs) have been proposed as a method to disentangle internal representations in language models, extracting features from superposition in an unsupervised manner (Templeton et al., 2024; Bricken et al., 2023; Cunningham et al., 2023). Notably, these features appear to be actionable: artificially activating them during inference can steer a model's output (Templeton et al., 2024; Makelov, 2024). Such methods have been successfully applied to language (Templeton et al., 2024; Cunningham et al., 2023; Gao et al., 2024), as well as vision and multimodal models (Gorton, 2024; Surkov et al., 2024), but biological and protein sequence models remain relatively unexplored (Simon & Zou, 2024; Adams et al., 2025).

Protein language models have been shown to encode structural, functional, and evolutionary information in their internal representations (Rives et al., 2019; Lin et al., 2023; Hayes et al., 2024). Interpretability methods for these models could reveal biological mechanisms, and support model debugging and editing for safety considerations. Additionally, model steering can be incorporated into sequence design pipelines.

The main contributions of this paper are:

• A trained sparse autoencoder (SAE) for the ESM-2 8M parameter model, along with potential interpretations of its latent components (sections 2.2, 3.1 and 3.3).

• A methodology for generating protein sequences by intervening on specific latents, demonstrating successful steering towards non-trivial features, such as zinc finger domains (section 3.4).

• A heuristic for selecting the model layer from which to extract representations using an intrinsic dimension estimator (section 3.2).

2 Background

2.1 Protein Language Models

Many advancements in Natural Language Processing have been successfully applied to biological sequence modeling. Transformer-based neural networks can be trained on protein sequences using the Masked Language Modeling (MLM) task, where each amino acid is treated as a token that can be randomly masked. The model learns to predict the masked tokens by minimizing the following loss function (Rives et al., 2019):

$$\mathcal{L}_{\text{MLM}} = \mathbb{E}_{x \sim X} \mathbb{E}_M \sum_{i \in M} -\log p(x_i | x_{/M}) \tag{1}$$

where x is a protein sequence, M is a set of masked indices and $p(x_i|x_{/M})$ is the probability assigned to the ground truth amino acid x_i given its sequence context.

Training on the Masked Language Modeling (MLM) task forces the network to learn dependencies between masked amino acids and their sequence context while simultaneously capturing various biological features present in the data. Embeddings extracted from these models have been shown to encode information about secondary structure, tertiary contacts (residue-residue interactions), function, remote evolutionary relationships, and factors relevant to predicting mutational effects (Rives et al., 2019; Elnaggar et al., 2021; Meier et al., 2021; Lin et al., 2023; Hayes et al., 2024). On the other hand, the attention mechanism appears to prioritize binding sites, with attention maps capturing information about residue-residue interactions. (Vig et al., 2020).

2.2 Sparse Autoencoders

The sparse autoencoders used for interpretability are simple, single-layer models trained on the activations of a larger language model. To disentangle network features, the hidden layer is made significantly larger than the original embeddings, creating an overcomplete basis. A sparsity constraint is then applied to ensure that only a few latent neurons are active at a time, making the SAE's hidden representation far more interpretable than standard language model components (Templeton et al., 2024; Bricken et al., 2023; Cunningham et al., 2023).

2.2.1 Architecture

The autoencoder is composed of an encoding and a decoding function, given by:

$$z = f_{enc}(x) = \text{ReLU}(W_{enc}(x - b_{dec}) + b_{enc}) \tag{2}$$

$$\check{x} = f_{dec}(z) = (W_{dec} \cdot z + b_{dec}) \tag{3}$$

Here f_{enc} is the encoder, that takes an embedded amino acid token $x \in \mathbb{R}^d$ from a given layer in the model and returns a latent $z \in \mathbb{R}^n_{\geq 0}$ with a hidden dimension n that is m times bigger that of the original vector (expansion factor). The decoder f_{dec} approximately reconstructs x given z, through the decoding matrix $W_{dec} \in \mathbb{R}^{n \times d}$ and the bias weight $b_{dec} \in \mathbb{R}^d$.

The loss function used for the training is a combination of the reconstruction error of the autoencoder $\mathcal{L}_{MSE}$ plus a sparsity constraint $\mathcal{L}_{L_1}$:

$$\mathcal{L}(x) = \mathcal{L}_{MSE} + \mathcal{L}_{L_1} = \sum_d (x_d - \check{x}_d)^2 + \lambda \sum_n z_n \tag{4}$$

While training, we renormalize the W_{dec} matrix to have unit norm after each backward pass. This is necessary to prevent that autoencoder latents become arbitrarily small and satisfy the L_1 constraint without actually being sparse.

3 Methods

3.1 Training Details

We use the ESM-2 family of models (Lin et al., 2023) as our base, extracting activations from the final output of the transformer block. We train on approximately 15k non-redundant protein sequences from SCOPe 2.08 (Fox et al., 2014). Further details on the architecture, training procedures, and hyperparameter selection can be found in section A.1 of the appendix.

3.2 Layer Selection

We adopt a principled strategy to select the layer from which we extract representations for the sparse autoencoder. The initial intuition, in line with earlier studies (Templeton et al., 2024; Gao et al., 2024), is to choose a mid-to-late layer, where the model is assumed to have developed abstract features but is not yet focused on the output reconstruction task. However, unlike these prior works, we move beyond mere intuition by incorporating a quantitative measure based on intrinsic dimension.

Specifically, we compute the intrinsic dimension of each layer's representations using the estimator proposed by (Facco et al., 2017), and then identify where this value plateaus. Previous research has shown that layers corresponding to local minima or plateaus in intrinsic dimension are where abstract information is most clearly encoded (Valeriani et al., 2024). Selecting a layer within this plateau increases the likelihood of capturing meaningful representations, providing a stronger foundation for interpretability and model steering.

3.3 Interpreting Autoencoder Latents

We extract protein annotations from the UniProt database (Uniprot, 2025) and convert them into binary labels for each amino acid in the sequence. We then compute the precision π and recall ρ of each latent component k in detecting a given feature ϕ.

Let A be the set of all amino acids and A_{ϕ^+} the set of amino acids that have been annotated with the feature ϕ. Considering a latent k to be active (k^+) for a given amino acid when its value z_k exceeds a certain threshold τ_z, we have:

$$\pi = P(\phi^+ | k^+) = \frac{\left|\left\{a \in A_{\phi^+} : z_k > \tau_z\right\}\right|}{\left|\left\{a \in A : z_k > \tau_z\right\}\right|} \tag{5}$$

$$\rho = P(k^+ | \phi^+) = \frac{\left|\left\{a \in A_{\phi^+} : z_k > \tau_z\right\}\right|}{\left|A_{\phi^+}\right|} \tag{6}$$

This gives us a value of precision and recall for each pair of k, ϕ. We consider a latent component to be associated with a specific feature if its precision or recall exceed a predefined threshold that we set to 0.80.